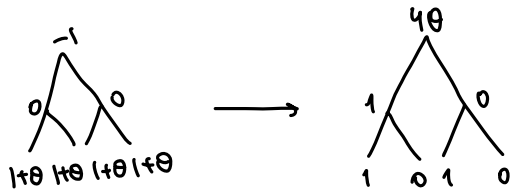
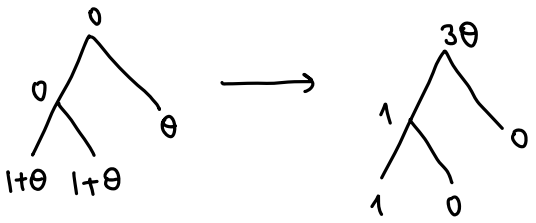
Algorithm to reduce trees

Must consider all the bottom options for the tree, prioritizing $\wedge\wedge$ final subtrees.

① 

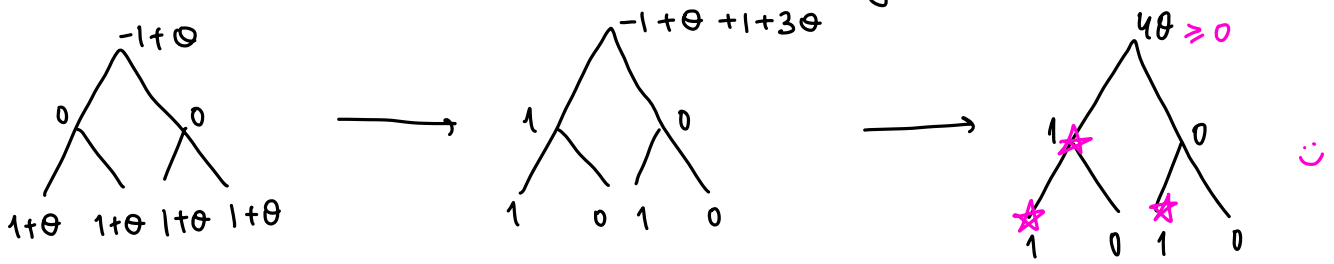
and we can always say that $4\theta \geq 1$ and keep 1 at the top

② 

this one is a problem if we want to claim the estimate for $s < -\frac{2}{3}$, so can look further up in the tree.

Options for root

- $F(1) = -2 - s = -2 + |s| = -1 - \theta$, only for $\wedge\wedge$ tree:



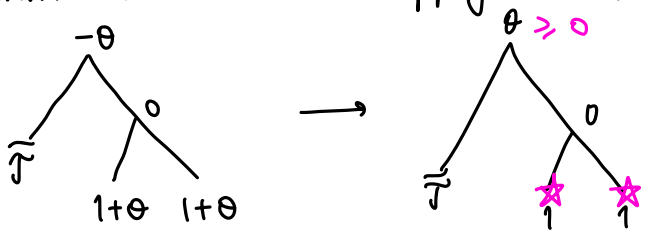
- $F(1) = -1 - s = -1 + |s| = -\theta$. The corresponding tree is always of the following form

(or swapping 10 and 11) where $\tilde{T} \in T_{j-2}$

To not fall in the previous case, we need $j-2 \geq 2$, so $j \geq 4$ then.

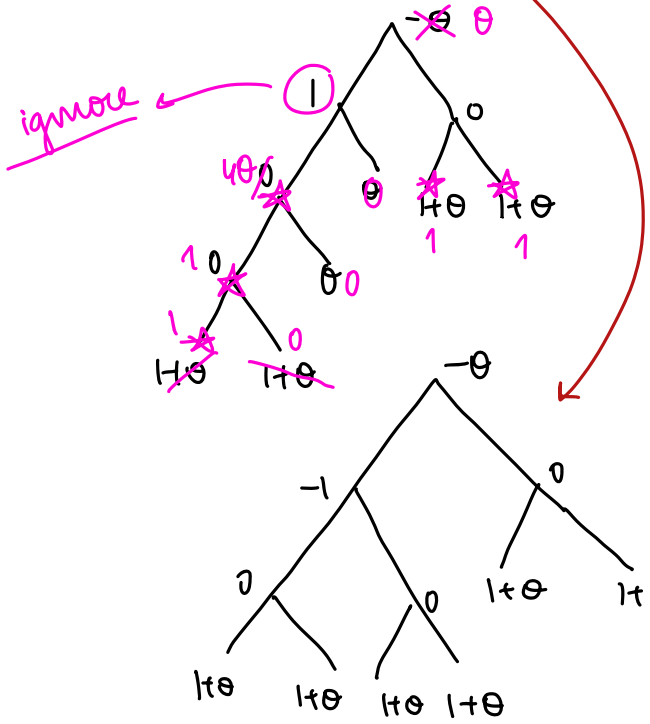
Consequently, we must have $F(10) \in \{0,1\}$ but never -1. not true! See ⚠

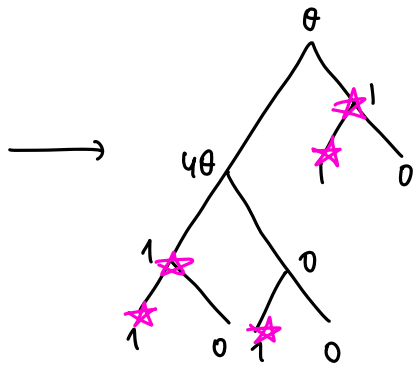
In the following case, we will handle a general tree with root 0, and the algorithm there should apply to $\tilde{T}$, then



i.e., we choose to integrate in 110 and 111, and apply the algorithm in the next step to $\tilde{T}$ (where we may have some extra powers which we are throwing away, in the case when $F(10) = 1$.

⚠ Must handle $\wedge\wedge$ separately





and $\theta > 0$ is enough

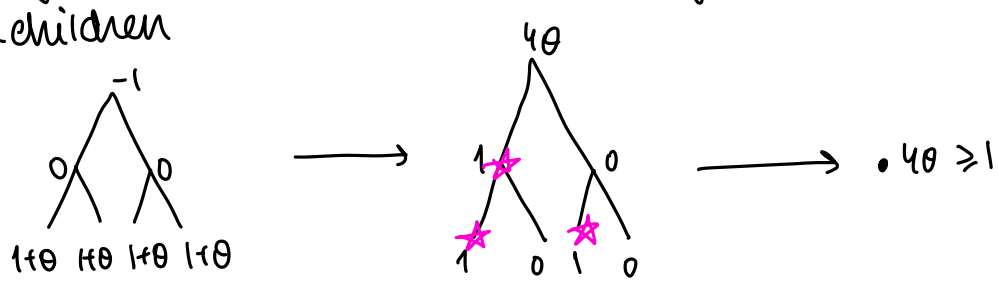
Thus, we can indeed assume that $F(10) \in \{0,1\}$, as this is the only choice of $\tilde{T}$ which will have a root with -1.

<u>case 3</u>: $F(1) = 0$

Here we develop an algorithm to handle all trees systematically.

<u>Step 1</u>: handle all -1's in the tree

The only nodes with -1 are final grandparents with 4 final grandchildren



→ Select 2 grandchildren which are not siblings, say $*00$ and $*10$, and 1 child, say $*0$, to integrate in

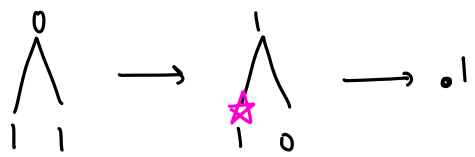
$$\int_{\substack{\xi_* = \xi_{*0} + \xi_{*1} \\ \xi_{*0} = \xi_{*00} + \xi_{*01} \\ \xi_{*1} = \xi_{*10} + \xi_{*11}}} \frac{|\xi_*|}{|\xi_{*00}\, \xi_{*01}\, \xi_{*10}\, \xi_{*11}|^{1+\theta}} \mathbb{1}_{A_*^c \cap A_{*0}^c \cap A_{*1}^c} \, d\xi_{*00}\, d\xi_{*10}\, d\xi_{*0}$$

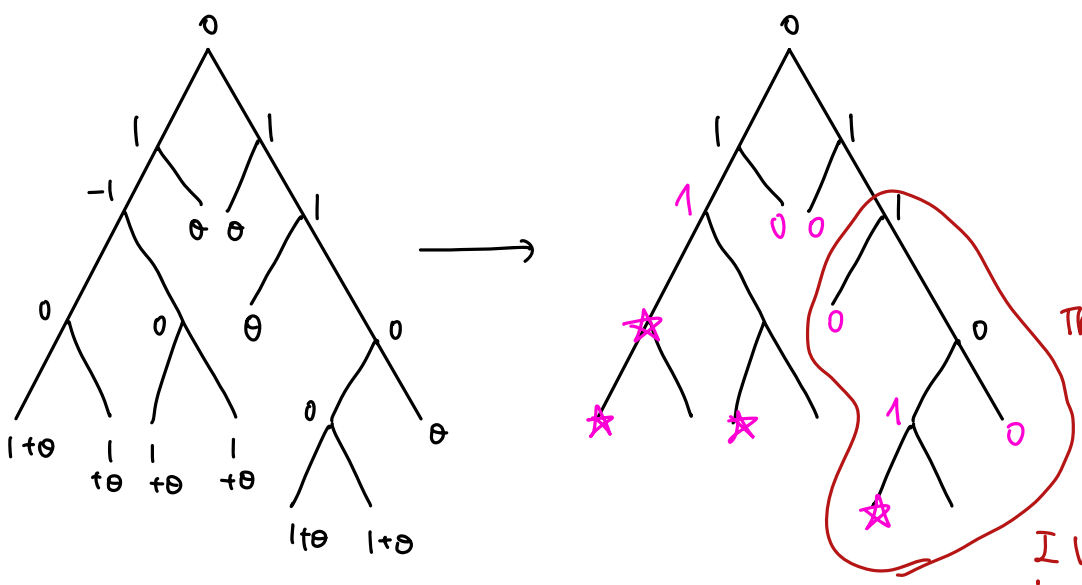
$$\lesssim \int_{\substack{\xi_* = \xi_{*0} + \xi_{*1} \\ \xi_{*0} = \xi_{*00} + \xi_{*01} \\ \xi_{*1} = \xi_{*10} + \xi_{*11}}} \frac{|\xi_*|^{1-1-4\theta+\varepsilon}}{|\xi_{*00}\,\xi_{*01}\,\xi_{*0}|^{1+\varepsilon}} \mathbb{1}_{A_*^c \cap A_{*0}^c \cap A_{*1}^c} \, d\xi_{*00}\, d\xi_{*10}\, d\xi_{*0}$$

$$\lesssim \int_{\substack{\xi_* = \xi_{*0} + \xi_{*1} \\ \xi_{*0} = \xi_{*00} + \xi_{*01} \\ \xi_{*1} = \xi_{*10} + \xi_{*11}}} \frac{1}{|\xi_{*00}\,\xi_{*01}\,\xi_{*0}|^{1+\varepsilon}\,|\xi_*|^{4\theta-\varepsilon}} \mathbb{1}_{A_*^c \cap A_{*0}^c \cap A_{*1}^c} \, d\xi_{*00}\, d\xi_{*10}\, d\xi_{*0}$$

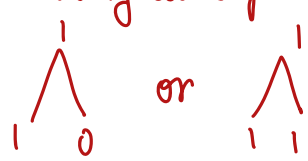
$$\lesssim \frac{1}{|\xi_*|^{1+\varepsilon}} \quad \text{for} \quad 4\theta - \varepsilon \geq 1 + \varepsilon$$

→ The above replaces a -1 with a 1 in the tree.
We also look at all $\overset{0}{\wedge}$ in the tree and replace by $\cdot 1$.
After this process, all nodes in the tree have 0 or 1 in them.
When we see a final tree





This tree structure is a problem too.
Need to look further.
I want to reduce to having always

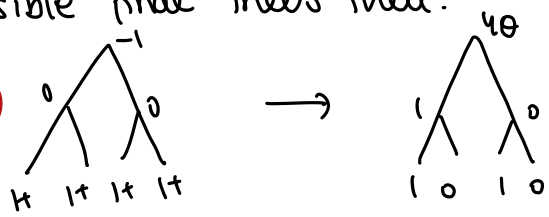


This means we want to look at both the parents nodes with $F(\cdot) = -1$ and $F(\cdot) = 0$.

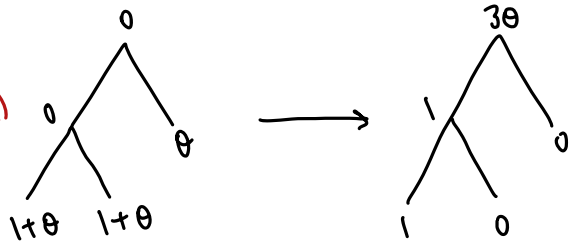
$F(\cdot) = -1$ ↓ final grandpa & final grandchildren

$F(\cdot) = 0$ ↓ final parents & final grandpa & final grandchildren

Possible final trees then:

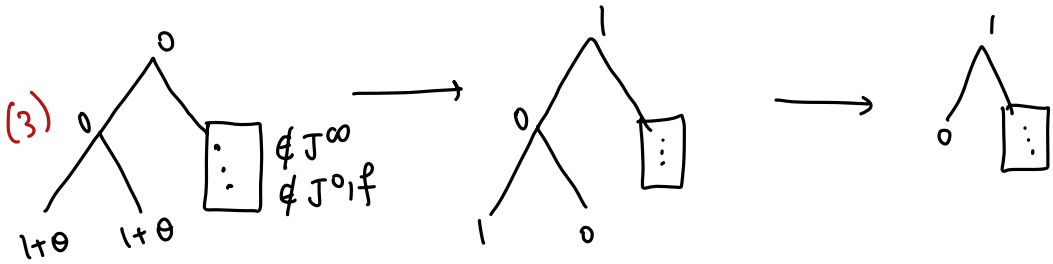
(1) 

$\cdot\, 4\theta \geq 1$ } only -1 nodes

(2) 

$\cdot\, 3\theta \geq 1$

$\boxed{\text{if we are happy with } s > -\frac{2}{3}}$

(3) 

After this reduction above, $\bullet \in \tilde{J}^0 \setminus \{1\} \Rightarrow F(\bullet) = 1$

$\bullet \in \tilde{J}^\infty \Rightarrow F(\cdot) \in \{0,1\}$

<u>claim</u>: never see $\bullet, s(\bullet) \in \tilde{J}^\infty$ with $F(\bullet) = F(s(\bullet)) = 0$.

The final nodes in $\tilde{J}$ are either final nodes from earlier tree $\bullet \in J^\infty$ with $s(\cdot) \notin J^\infty$, for which $F(\cdot) = 0$, or newly created final nodes by (1)–(3) above.

→ (1) creates a new final node with $F(\cdot) = 1$ ☺

→ (2) also creates a new final node with $F(\cdot) = 1$ ☺

→ (3) creates a final node with $F(\cdot) = 0$, but its sibling is not final in original tree (so not immediately 0). If $s(\cdot)$ becomes final in $\tilde{J}$ it must be from (1) or (2). (2) does not apply because $s(\cdot) \notin J^\infty$ by assumption. If (1) applies, then $\tilde{F}(s(\cdot)) = 1$ in new tree, as intended!

Therefore, we always have $\underset{1\ 1}{\wedge}$ or $\underset{0\ 1}{\wedge}$ or $\underset{1\ 0}{\wedge}$ in final chevrons in $\tilde{J}$. $\square$

<u>Claim</u>: all parent nodes in new tree have $F(\cdot) = 1$, except for the root.

In the original tree, for $\cdot \in J^{\infty}$, we had

$$F(\cdot) = \begin{cases} 0, & \cdot = 1 \\ -1, & \cdot 0, \cdot 1 \in J^{0,f} \\ 0, & \cdot 0 \in J^{0,f} \text{ or } \cdot 1 \in J^{0,f} \text{ exactly (as in exclusive or)} \\ 0, & \cdot \in J^{0,f} \\ 1, & \cdot, \cdot 0, \cdot 1 \notin J^{0,f} \end{cases}$$

The process of (1) - (3) removes the following parent nodes from $\mathcal{T}$:

→ $\cdot \in J^{0,f}$ are either not in $\tilde{\mathcal{T}}$ (1)-(2), or are in $\tilde{\mathcal{T}}^{\infty}$ (3)

↑ none of those are in $\tilde{\mathcal{T}}^{0}$!

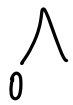
→ $\cdot \in J^{0,f}$ with $\cdot 0, \cdot 1 \in J^{0,f}$ are all turned into final nodes in $\tilde{\mathcal{T}}$ (1)

↑ none of these are in $\tilde{\mathcal{T}}^{0}$!

→ Also, from (3), for $\cdot \in J^{0}$ with $|\{\cdot 0, \cdot 1\} \cap J^{0,f}| = 1$ and $\cdot 0, \cdot 1 \notin J^{\infty}$, $F(\cdot) = 0$ but $\tilde{F}(\cdot) = 1$ in new tree!

Thus, for $\cdot \in \tilde{\mathcal{T}}^{0}$, we have

$$F(\cdot, \tilde{\mathcal{T}}) = \begin{cases} 0, & \cdot = 1 \\ 1, & |\{\cdot 0, \cdot 1\} \cap J^{0,f}| = 1 \\ 1, & \cdot, \cdot 0, \cdot 1 \notin J^{0,f} \end{cases} = \begin{cases} 0, & \cdot = 1 \\ 1, & \text{else} \end{cases}$$

Combining the 2 claims above, we get a new tree where all parent nodes (except the root) have power 1, all final nodes have 0 or 1, and if we see  in final subtree, the sibling must have 1.

We now repeat the following step until we reach the top of the tree:

→ if tree or then integrate in Σ_{10}

→ if not final tree of type above, find final subtree Λ.

- if integrate in $*0$ and collapse to $*^1$
- if symmetric
- if integrate in $*0$ and collapse to $*1$

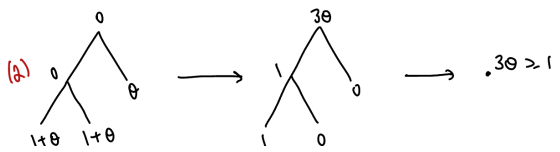
Note that the step above preserves the structure of the new tree with at least one node in each pair of final nodes with weight 1 and all parents except root with weight 1.

$\boxtimes$

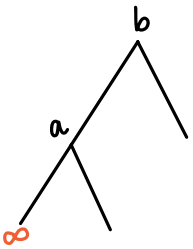
Improving algorithm to go beyond $s > -\frac{2}{3}$

($s > -\frac{3}{4}$ would be good)

This restriction came from the initial step

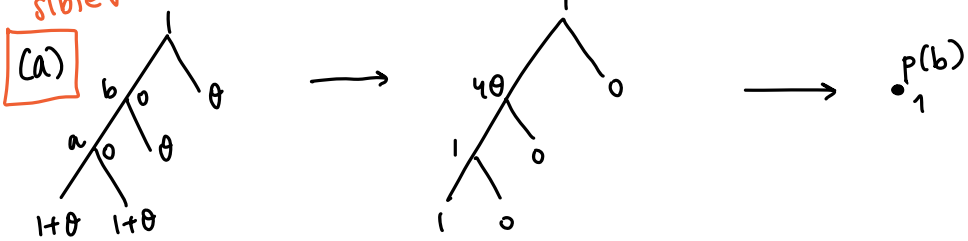


If we look further up in the tree, we should be able to avoid this.



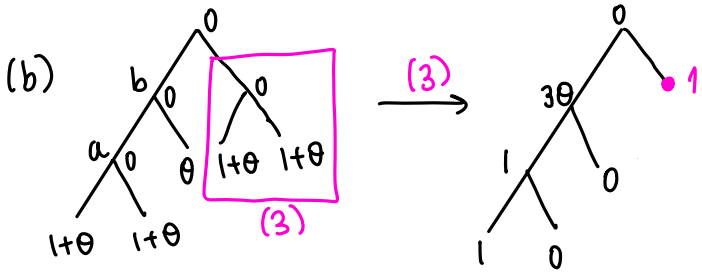
want to know what are the options for $p(b)$ and $s(b)$.

$s(b) \in \mathcal{T}^{\infty}$



with $4\theta \geq 1$:)

If root of subtree is root of $\mathcal{T}$, then same.

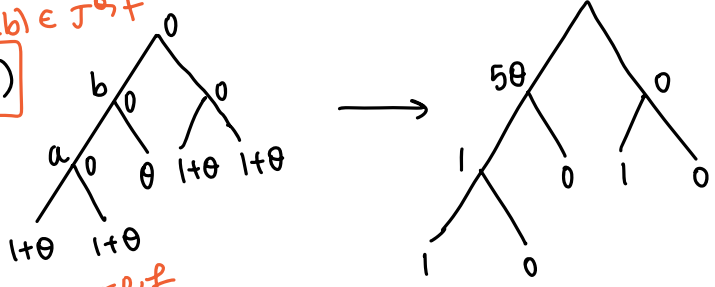


⚠ do not want to do (3) exactly.

Instead

$s(b) \in J^{0,f}$

(b)



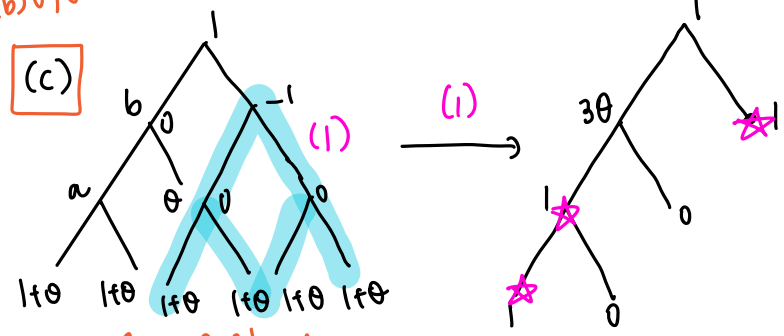
$\longrightarrow$ $\bullet^{1}_{p(b)}$ for $5\theta \geq 1$

This means we want to at least change (3) to $\boxed{\cdot} \neq \wedge$ as well

Now $\boxed{\cdot} \notin \{\bullet, \wedge, \lambda, \wedge\}$

$s(b)0, s(b)1 \in J^{0,f}$

(c)

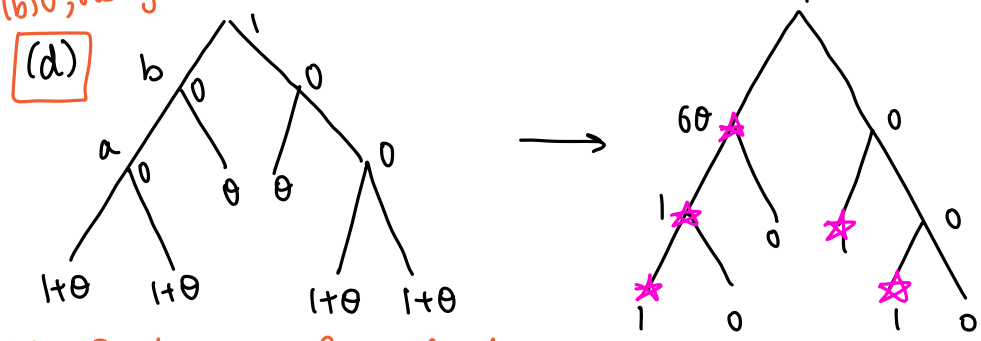


$\longrightarrow$ $\bullet_{1}$

so no changes needed, just using (1) is enough.

$|\{s(b)0, s(b)1\} \cap J^{0,f}| = 1$

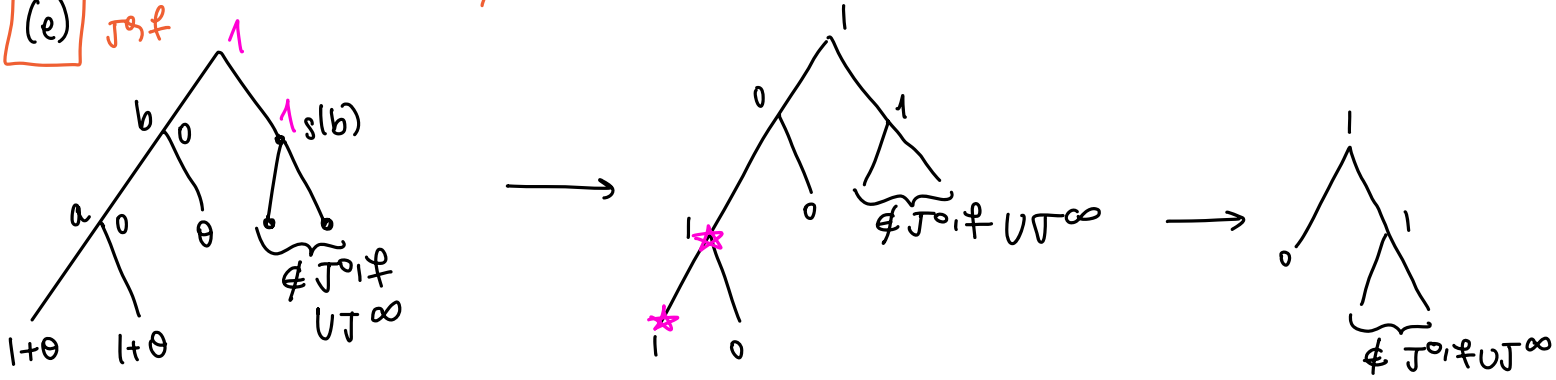
(d)



$\longrightarrow$ $\bullet^{1}_{p(b)}$ for $6\theta \geq 1$

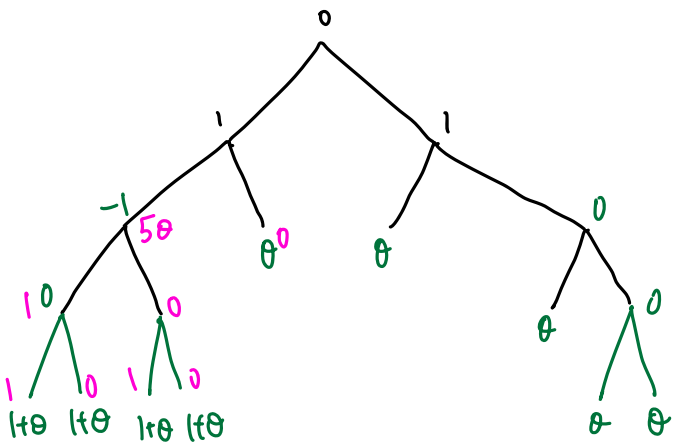
$s(b) \in J^{0}_{1}$, $\{s(b)0, s(b)1\} \cap J^{0,f} = \emptyset$

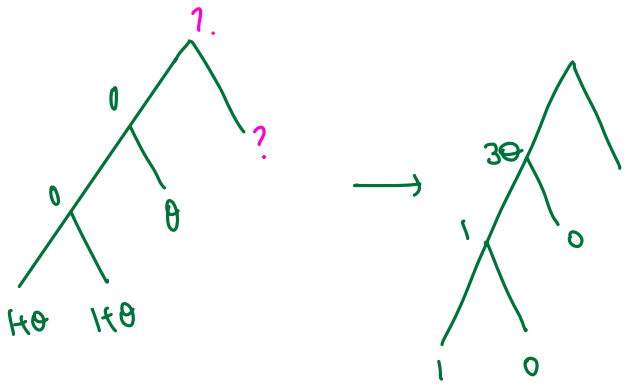
(e) $J^{0,f}$



- This covers all cases and the resulting tree is of the same type
- can write both versions of the algorithm. Simpler one and the modification to get the intended result.

Example of tree where you can never actually do $\theta \geq \frac{1}{j+1}$ and must take it larger

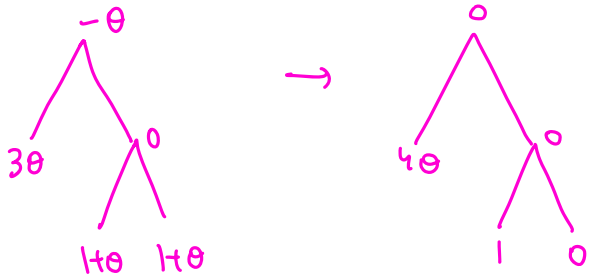




(1)

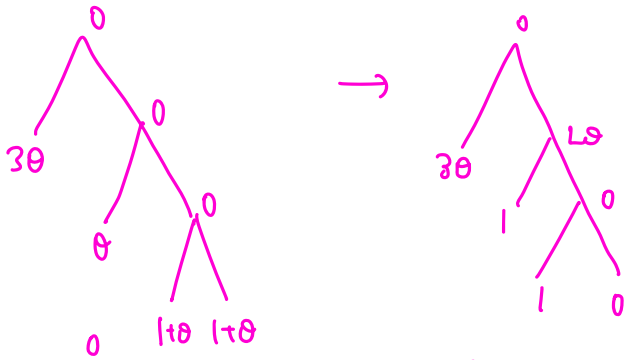
$4\theta \geq 1$

(2)



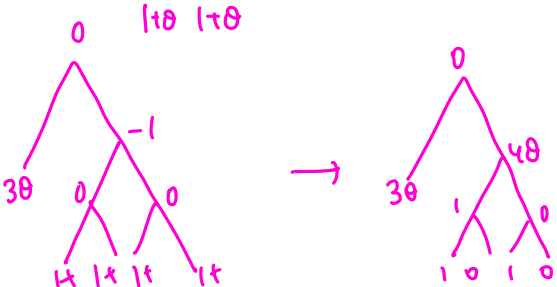
$4\theta \geq 1$

(3)



$5\theta \geq 1$

(4)



(5)

and all reductions under will always give a 1 and we will not need 3θ.

⚠ claim 4θ is the best we can do, because there will be an explicit tree that requires it. To do better systematically is not clear.